SIKSHA 'O' ANUSANDHAN
(Deemed to be University)
Faculty of Engineering & Technology (ITER)
Department of Computer Science and Engineering

Project Proposal Form

FINAL YEAR RESEARCH PROJECT-2026

SECTION: 26 **GROUP NO: 04**

PROJECT TITLE: AI-Based Early Disease Risk Prediction System Using Machine Learning

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PROJECT ABSTRACT:

The primary aim of this project is to design and develop an intelligent machine learning based system capable of predicting the early risk of chronic diseases such as diabetes and heart disease using patient health parameters. Early identification of disease risk enables timely preventive measures, reducing disease progression and associated healthcare burden. The objective is to build a reliable predictive model by analyzing clinical datasets and comparing multiple machine learning algorithms to determine the most accurate and clinically relevant approach for early risk assessment.

The problem addressed in this project is the lack of accessible and data-driven tools for preliminary disease risk screening, especially in resource-limited settings where frequent medical checkups may not be feasible. Many chronic diseases remain undetected until advanced stages due to the absence of early warning systems. This project proposes a computational decision-support solution that can estimate disease probability using routine health indicators such as age, body mass index, glucose level, blood pressure, and cholesterol.

Technically, the system will utilize publicly available medical datasets and implement data preprocessing, feature selection, and supervised machine learning models including Logistic Regression, Random Forest, and Gradient Boosting. Model performance will be evaluated using metrics such as accuracy, precision, recall, and ROC-AUC. The final system will be deployed through a user-friendly web interface that accepts patient parameters and outputs disease risk probability along with risk level categorization (low, medium, high).

The expected benefits of this project include promoting preventive healthcare awareness, assisting preliminary screening in rural or underserved areas, and supporting clinicians with data-driven insights. Socially, the system contributes toward early intervention, reduced healthcare costs, and improved public health outcomes by enabling individuals to recognize potential disease risks at an early stage

(1) SOFTWARE, HARDWARE OR METHODS/ALGORITHMS SPECIFICATIONS:

The proposed system will be developed using Python as the primary programming language due to its extensive support for machine learning and data analysis. Data preprocessing, analysis, and visualization will be performed using libraries such as NumPy, Pandas, Matplotlib, and Seaborn. Machine learning models including Logistic Regression, Random Forest, Support Vector Machine, and Gradient Boosting (XGBoost) will be implemented using the Scikit-learn and XGBoost frameworks..

Publicly available medical datasets such as the Pima Indians Diabetes dataset and UCI Heart Disease dataset will be utilized for training and evaluation. The system will follow a supervised learning approach involving data cleaning, feature selection, model training, validation, and performance comparison. Evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC will be used to determine model effectiveness.

The trained predictive models will be deployed through a web-based interface developed using the Streamlit framework, enabling users to input health parameters and obtain disease risk predictions with categorized risk levels

(2) NAME, REG. NO AND SIGNATURE OF GROUP MEMBERS:

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(3) APPROVAL STATUS (To be filled in by the Section Coordinator of FRP):

Bhanu Pratap Rajak	....	....
Project Supervisor	**Section Coordinator, FRP**	**FRP Coordinator**